Kevin Lee

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Embedded Systems Engineer with 3+ years of experience developing production C firmware for automotive traction inverters and recent C++ drive-by-wire development for autonomous vehicles. Passionate about hands-on embedded development, hardware/software debugging, and performance optimization, with ongoing personal MCU projects. Experienced in real-time control, CAN communication, and automated HiL verification.

WORK EXPERIENCE

Hardware Engineer Summer Intern (Vehicle Embedded Software), WeRide Corp, San Jose **Jun 2026 – Sep 2026**

- Developed and maintained drive-by-wire embedded software for vehicle actuation, integration, and system-level debugging.
- Developed and deployed a nonlinear low-speed brake control in C++, eliminating hunting and near-zero torque overshoot.
- Diagnosed vehicle control issues using CAN logs and road tests, driving firmware fixes.
- Validated embedded control software through vehicle testing and issue reproduction.

Senior Firmware Engineer, Delta Electronics Inc. **Oct 2021 - May 2025**

- Developed production embedded C firmware for EV traction inverters, supporting releases, EOL, and customer RMA.
- Debugged MCU firmware and hardware using Lauterbach, schematics, and oscilloscopes, tracing registers, interrupts, SPI/DMA/CAN bus data flow, and circuit signals.
- Debugged and integrated MCU bootloader and application firmware, verifying software jumps, interrupt behavior and peripheral initialization using Lauterbach/Trace32.
- Optimized real-time embedded software execution and CPU utilization, reducing runtime load from 55% to 35%.
- Developed and automated unit, integration, and qualification testing using VectorCAST and dSPACE HiL, reducing verification effort by 85%.
- Led software verification for ASPICE Level 2/3 program, ensuring compliance with requirements and verification processes.
- Diagnosed inverter-level issues and supported customer RMA with global engineering and customer teams across the U.S., Europe, and Asia.

Project Lead, Advanced Power R&D Center, National Taiwan University **Aug 2019 - Jul 2021**

- Led a 6-person team to successfully complete two automotive cooperative research projects.
- Developed an EV truck by-wire HIL platform, supporting subsystem development and powertrain integration.
- Designed and implemented an 8kW plug-In range-extended hybrid powertrain for a scooter prototype.
- Built STM32-based VCU embedded software for energy-management control and CAN-FD communication.
- Achieved 8.5% CO_2 reduction compared to a conventional ICE powertrain using the hybrid prototype.

EDUCATION

University of California, Irvine, CA **Sep 2025 – Expected Dec 2026**
Master's Degree, Embedded and Cyber-Physical Systems

National Taiwan University, Taipei, Taiwan **Aug 2019 - Jul 2021**
Master's Degree, Mechanical Engineering

National Cheng Kung University, Tainan, Taiwan **Aug 2014 - Jul 2018**
Bachelor's Degree, Mechanical Engineering

SKILLS

- Programming: C, C++, Python, Simulink, FreeRTOS, RTA-OS
- Development Platforms: Git, Jenkins, IBM Rhapsody, IBM Rational, Enterprise Architect
- Embedded Platforms: STM32, ARM Cortex-M (STM32U545/M33, MAX32690/M4F), ESP32, Infineon AURIX, Arduino, Raspberry Pi
- Validation Platforms and Tools: Lauterbach, Trace32, PCAN, CANalyzer, CANoe, VectorCAST, HelixQAC, dSPACE
- Standards/Frameworks: MISRA C, ASPICE, ISO26262 Software Verification, HiL, MiL
- Communication Protocols: CAN BUS, CAN-FD, I2C, I2S, SPI, UART, BLE